Predicting Decision-Time Reliability of Equity Market Risk under Delayed Inputs and Non-Stationary Conditions

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Abstract

Value-at-Risk (VaR) models are standard in automated risk control, but operational pipelines consume risk reports with a one-day reporting lag while market regimes shift faster than the estimator adapts. This mismatch creates a decision-time gap in which a nominally current risk estimate is stale. We formalize the Decision-Time Reliability problem and propose a predictive monitoring layer that estimates next-day failure probability using only decision-time information. Leave-one-out ablation and SHAP analysis independently confirm that risk-volatility divergence is the dominant predictive signal, dominant by a 2x margin in mean absolute SHAP value, and removing noisy features improves XGBoost ROC-AUC from 0.642 to 0.704. We compare four models (volatility threshold, logistic regression, pruned XGBoost, and our novel Regime-Conditional Reliability Estimator, RCRE) across six global equity markets: S&P 500, FTSE 100, Nikkei 225, DAX, Hang Seng, and EM ETF. The volatility-threshold heuristic achieves near-random discrimination (AUC ≈0.50) on every asset. A reliability-gated three-level exposure policy reduces Expected Shortfall Breach (ESB) by 42 to 67 percent consistently across all six assets. RCRE achieves the best tail severity reduction (56 percent ESB reduction on S&P 500) via soft Gaussian regime mixing. The COVID-19 volatility shock serves as a fully held-out stress test, with the monitor triggering de-risking windows during the acute mid-March crash window.

1 INTRODUCTION

Risk limits are enforced using model-derived signals that are rarely synchronized with the market state at decision time. VaR and ES estimates are produced on rolling windows and published with delay, while volatility and tail behaviour can shift abruptly. We treat this as a systems problem: rather than redesigning the base estimator, we learn a reliability monitor that predicts when the reported risk will fail to bound next-day losses under operational constraints.

Operational requirements. Latency is explicit: the consumed risk signal at time t is computed using information only up to t−L where $L \in \mathbb{N}$ denotes the operational lag. Temporal causality is strict: the baseline risk report at time t is $\mathcal{F}(t-1)$-measurable, while reliability features are $\mathcal{F}(t)$-measurable. All labels are realized strictly after decision time.

1.1 Formal Problem Statement

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with discrete-time filtration $\{\mathcal{F}(t)\}_{t+}\mathbb{N}$. Let r(t) denote log-returns and $\hat{R}(t)$ a risk estimate (VaR at level α).

Definition 1 (Operational Latency Gap). The risk signal consumed at time t is computed under lag L ≥ 1. The estimate $\hat{R}(t|t-L)$ denotes the value available at t computed using $\mathcal{F}(t-L)$ only:

$$\hat{R}(t \mid t-L) = g(\, r(1:t-L)\,;\ \theta(t-L)\,)$$

Definition 2 (Reliability Failure). Let $L(t) := -r(t)$ denote realized loss and κ > 0 an operational safety buffer. The reliability failure indicator is:

$$z(t) = 1\{\, -r(t) > \kappa \cdot \hat{R}(t \mid t-L)\,\}$$

A failure event z(t) = 1 indicates the delayed risk estimate is no longer conservative under the conditional distribution due to non-stationary shifts. We focus on the operationally common case L = 1.

Table 1: Notation

Symbol	Meaning
P(t)	Adjusted close price
r(t)	Log return ln(P(t) / P(t−1))
L(t)	Realized loss: −r(t)
$\hat{R}(t\|t-L)$	Risk estimate at t, computed using $\mathcal{F}(t-L)$
z(t)	Failure indicator: $\mathbb{1}\{ L(t) > \kappa \cdot \hat{R}(t\|t-L) \}$
p(t) / s(t)	Raw / calibrated failure probability $P(z(t+1)=1 \mid \mathcal{F}(t))$
π(t)	Exposure multiplier in {1, 0.5, 0} from gating policy

All symbols used throughout the paper. Notation follows standard probability space conventions.

1.2 Motivation

Traditional backtesting evaluates violations over a window [t−N, t]. While useful for retrospective validation, it is inherently ex-post and fails to indicate whether $\hat{R}(t|t-1)$ is actionable at decision time. We build a Reliability Prediction Layer that estimates:

$$p(t) = \mathbb{P}(\, z(t+1) = 1 \mid \mathcal{F}(t)\,)$$

to proactively trigger conservative controls before violations occur. Rather than replacing the legacy estimator g, our layer monitors the divergence between g's output and the current market state.

1.3 Contributions

- **Decision-Time Reliability Formalism.** We formalize risk reliability under delayed inputs with strict filtration measurability, distinguishing ex-post backtesting from forward conditional failure prediction.
- **Feature Ablation and Signal Identification.** Leave-one-out ablation and SHAP TreeExplainer independently identify risk-volatility divergence as the dominant signal (ablation ΔAUC = −0.079; mean|SHAP| = 1.114, 2x the next feature). Removing noisy features improves XGBoost AUC by +0.062.
- **RCRE: Regime-Conditional Reliability Estimator.** A soft-mixing architecture that trains regime-specific XGBoost classifiers and blends predictions via a Gaussian recency kernel, addressing non-stationarity within the monitor itself.
- **Multi-Asset Validation.** Six-asset evaluation across US, European, and Asian markets. ESB reduction of 42 to 67 percent holds consistently, establishing cross-market generalisability.
- **Aperture Effect.** Reliability gating converts rare catastrophic breaches into more frequent but materially smaller violations, a structural trade-off that traditional backtesting fails to capture.

2 RELATED WORK

2.1 Backtesting and Coverage Rigor

VaR and ES validation relies on coverage and independence tests. Modern standards emphasise the elicitability properties of joint backtests [10], designed for long-horizon retrospective validation [6]. We instead treat failure as a predictable state variable, targeting point-in-time actionability.

2.2 ML in Non-Stationary Time Series

GBDTs and Transformers have improved tail-relevant prediction in financial time series [1, 13]. Non-stationarity remains a central failure mode, motivating shift-aware techniques such as Dish-TS [5]. We shift the objective from point estimation to predicting the reliability of a delayed legacy risk report under operational latency.

2.3 Decision-Centric and Safety-Critical ML

Safety-critical ML increasingly emphasises prediction of output trust for downstream actions [2]. Probability calibration via isotonic regression [12] is a prerequisite for threshold-based policy. SHAP [9] provides interpretable feature attribution grounded in game theory. Our work adapts these methods to market risk, aligning with ECB [4] guidance on continuous model monitoring.

3 METHODOLOGY

3.1 Data and Information Timing

We use daily log-returns of six equity indices from January 2000 through end-2024 (approximately 6,300 days per asset). Log-returns are computed as:

$$r(t) := ln\frac{P(t)}{P(t-1)}$$

The current-day features use $\mathcal{F}$(t); the baseline risk report $\hat{R}(t|t-1)$ uses data ending at t−1. Realized losses are used only for labelling and evaluation. Chronological split: 60 percent train, 20 percent validation, 20 percent test, with no reshuffling.

3.2 Baseline Risk Estimator

Rolling Historical Simulation VaR at α = 0.99 over a 252-day window on delayed returns:

$$\hat{R}(t|t-1) = Q_{n\cdot nn}\,[\,L^d(t-252), \ldots, L^d(t-1)\,]$$

EWMA volatility proxy with λ = 0.94:

$$\hat{\sigma}^2(t) = \lambda \cdot \hat{\sigma}^2(t-1) + (1-\lambda) \cdot [\,r^d(t-1)\,]^2$$

The 252-day lookback creates systematic staleness during abrupt regime shifts. This is the core failure mode the reliability layer targets.

3.3 Failure Label

Failure is defined as a breach of baseline VaR scaled by γ = 0.8:

$$\tilde{y}(t) := 1\{\,L(t) > \gamma \cdot \hat{R}(t|t-1)\,\}$$

The supervised target is the next-day outcome: y(t) := ỹ(t+1), a strict next-day forward label. Using γ = 0.8 flags losses exceeding 80 percent of VaR, giving a 3.15 percent positive rate. Class imbalance is handled via scale_pos_weight = (#neg) / (#pos).

3.4 Feature Map and Ablation

We initially construct a 7-feature candidate set, then perform leave-one-out ablation to identify the optimal subset. All features are $\mathcal{F}$(t)-measurable by construction. Table 2 reports all candidate features with ablation results. Two features are removed: kurtosis (removing it improves AUC by +0.071) and exceed_count (removing it improves AUC by +0.015). The final feature vector is:

$$\varphi(t) = [\,\hat{\sigma}(t),\ \Delta\hat{\sigma}(t),\ \hat{R}(t|t-1)-\hat{\sigma}(t),\ L(t)-\hat{R}(t|t-1),\ |L(t)|\,]$$

Table 2: Feature Ablation Results

Positive ΔAUC means removal improves the model (noise feature). The pruned 5-feature model achieves AUC 0.704 vs 0.642 for the full 7-feature set.

Feature	Definition	ΔAUC if removed	Retained	Reason
Risk-Vol Divergence ($\hat{R}-\hat{\sigma}$)	Estimator staleness	−0.079 (dominant)	Yes	Core signal
Loss-Risk Divergence ($L-\hat{R}$)	Instant breach size	−0.021	Yes	Breach signal
Absolute Loss $\lvert L(t)\rvert$	Raw market stress	−0.012	Yes	Stress proxy
EWMA Volatility ($\hat{\sigma}$)	Current vol regime	−0.006	Yes	Regime state
Volatility Momentum ($\Delta\hat{\sigma}$)	Vol rate of change	−0.006	Yes	Trend signal
Exceed Count	20-day breach count	+0.015	No	Noise
Excess Kurtosis (60-day)	Tail heaviness	+0.071	No	Noise
Pruned 5-feature model (final)		AUC: 0.704 (+0.062)	Final	

3.5 Reliability Model and Isotonic Calibration

We train an XGBoost binary classifier $f(\theta)$ to map features to a one-step-ahead failure probability:

$$p(t) := f\theta(\varphi(t)) \approx \mathbb{P}(z(t+1) = 1 \mid \mathcal{F}(t))$$

Since the gating policy relies on probability thresholds, we calibrate p(t) using isotonic regression fit on the validation block only:

$$\hat{s}(t) := g(p(t)), \quad g \text{ fitted by PAVA on validation data only}$$

Isotonic calibration is non-parametric and makes no assumption about the shape of the score-to-probability mapping. The calibrated score s(t) is used for all downstream gating decisions.

3.6 RCRE: Regime-Conditional Reliability Estimator

A single classifier trained on pooled data conflates calm-market and stress-market signal. RCRE addresses non-stationarity via three stages.

Stage 1: Regime Detection. CUSUM change-point detection [8] on log-volatility, followed by k-means clustering (k=3) of local volatility statistics. Labels are mapped to Calm (0), Transitional (1), and Crisis (2) by mean volatility. Centroids are fitted on training data only and applied forward, preserving strict temporal causality:

$$regime(t) = argmin_k \parallel S(\varphi(t)) - c_k \parallel_2$$

Stage 2: Regime-Specific Classifiers. Separate XGBoost classifiers $f(\theta_k)$ are trained per regime, each with its own isotonic calibrator g(k) fitted on regime-specific validation data:

$$\hat{s}_k(\varphi(t)) = g_k(f\{\theta_k\}(\varphi(t)))$$

Stage 3: Soft Gaussian Mixing. Hard regime assignment creates probability discontinuities at transition boundaries. RCRE computes soft weights w(t) in the probability simplex using a backward Gaussian kernel:

$$w_k(t) = \frac{\sum_{i=0}^{W-1} \exp(\frac{-i^2}{2r^2}) * 1[regime(t-1) = k]}{Z}$$

where W = 10 (mixing window) and τ = 3.0 (bandwidth). The final prediction is:

$$\hat{p}RCRE(t) = \sum_k w_k(t) \cdot \hat{s}_k(\varphi(t))$$

Soft mixing ensures smooth probability transitions at regime boundaries, avoiding cliff-edge gating behaviour. This is the key architectural distinction from plain XGBoost.

3.7 Comparison Baselines

- **Volatility Threshold (VT).** Binary flag: $\mathbb{1}\{\hat{\sigma}(t) \geq q_{90}(\hat{\sigma})\}$, threshold set on training data. The most natural operational heuristic.
- **Logistic Regression (LR).** L2-regularised (C=0.1) on the same 5 features with isotonic calibration. Captures only linear signal in the feature space.
- **XGBoost (5-feature, post-ablation).** Same hyperparameters as RCRE per-regime models, trained on pooled data. Represents the single-model non-linear baseline.

3.8 Reliability-Gated Decision Policy

We define a three-level exposure multiplier $\pi(t) \in \{1, 0.5, 0\}$ with thresholds $\tau_low = 0.019$ (70th score percentile) and $\tau_high = 0.064$ (90th score percentile), set on validation data and fixed for test evaluation:

$$\pi(t) = 1 \quad \text{if } \hat{p}RCRE(t) < \tau_{low}$$

$$\pi(t) = 0.5 \quad \text{if } \tau_{low} \leq \hat{p}RCRE(t) < \tau_{high}$$

$$\pi(t) = 0 \quad \text{if } \hat{p}RCRE(t) \geq \tau_{high}$$

The gated risk limit is $\Lambda(t) = \gamma \cdot \pi(t) \cdot \hat{R}(t|t-1)$, and the gated PnL stream applies $\pi(t)$ to the next-day return.

Algorithm 1: RCRE (Regime-Conditional Reliability Estimator)

Input: Feature matrix $\varphi(1:T)$, labels $y(1:T)$, number of regimes $\kappa = 3$, mixing window $W = 10$, kernel bandwidth $\tau = 3.0$

1. Fit a CUSUM change-point detector on log-volatility of training returns
2. Fit k-means (K = 3) on local volatility statistics; record frozen centroids c_1, c_2, c_3
3. Assign regime labels to all training observations:

$$regime(t) = argmin_k \, \| S(\varphi(t)) - c_k \|_2$$

4. For each regime $k \in \{0, 1, 2\}$:

   Train XGBoost classifier $f\theta_k$ on the subset $\{ (\varphi(t), y(t)) : regime(t) = k \}$

5. Fit isotonic calibrator g_k on the regime-k validation subset
6. Assign current regime using frozen centroids:

$$regime(t) = argmin_k \, \| S(\varphi(t)) - c_k \|_2$$

7. Compute backward Gaussian mixing weights for each regime k:

$$w_k(t) = \frac{\sum_{i=0}^{W-1} \exp(\frac{-i^2}{2\tau^2}) * 1[regime(t-1) = k]}{Z}$$

8. Normalize weights to the probability simplex:

$$w_k(t) \leftarrow \frac{w_k(t)}{\Sigma_j \, w_j(t)}$$

9. Compute calibrated score from each regime-specific model:

$$s_k = g_k(f\theta_k(\varphi(t))) \; \forall k \in \{0, 1, 2\}$$

10. Compute soft-mixed final prediction:

$$\hat{p}RCRE(t) = \sum_k w_k(t) \cdot s_k$$

Output: Calibrated failure probability $\hat{p}_RCRE(t) \approx \mathbb{P}(z(t+1) = 1 \mid \mathcal{F}(t))$

4 EXPERIMENTAL RESULTS

4.1 Evaluation Metrics

- **Discrimination:** ROC-AUC and PR-AUC on the chronological test block.
- **Calibration:** Brier score

$$BS = (\frac{1}{T})(\sum_{T}^{t=1} (s(t) - y(t))^2$$

- **Violation Rate (VR):**

$$VR := (\frac{1}{T}\sum_{t-1}^{T} 1\{ L(t) > \Lambda(t) \},$$

  where $\Lambda(t) = \gamma \cdot \pi(t) \cdot \hat{R}(t|t-1)$.
- **Expected Shortfall Breach (ESB):**

$$ESB := E[L(t) \mid L(t) > \Lambda(t)].$$

  Primary tail severity metric.

4.2 Discrimination: S&P 500

Table 3 reports test-block results on S&P 500. The volatility threshold achieves AUC 0.508, near random, confirming that raw volatility level is insufficient. The 7-feature XGBoost achieves AUC 0.642; ablation-guided pruning to 5 features improves this to 0.704 (+0.062), demonstrating that feature quality matters more than feature quantity. RCRE (AUC 0.703) outperforms the 7-feature XGBoost and achieves the best Brier score (0.0323), indicating superior calibration. Logistic regression (AUC 0.755) remains competitive, reflecting the strong linear component of the risk-volatility divergence feature.

Table 3: S&P 500 Test Block Discrimination and Caliberation

Model	ROC-AUC	PR-AUC	Brier Score	Calibrated
Volatility Threshold	0.508	n/a	n/a	No
Logistic Regression	0.755	n/a	n/a	Yes
XGBoost 7-feat (pre-ablation)	0.642	0.056	0.034	Yes
XGBoost 5-feat (post-ablation)	0.704	0.130	0.0324	Yes
RCRE (ours)	0.703	0.125	0.0323	Yes

Ablation improves XGBoost AUC by +0.062. RCRE achieves best Brier score. LR competitive due to strong linear signal in risk-volatility divergence.

4.3 SHAP Interpretability Analysis

To validate ablation findings via an independent interpretability method, we apply SHAP TreeExplainer [9] to the trained XGBoost model on the held-out test set. SHAP decomposes each prediction into per-feature contributions grounded in cooperative game theory. Crucially, SHAP and leave-one-out ablation are methodologically independent: one measures prediction decomposition, the other measures performance degradation. Agreement between them constitutes strong evidence that the identified signal is real.

Table 4 reports mean absolute SHAP values per feature. Risk-volatility divergence is the dominant driver with mean|SHAP| = 1.114, approximately 2x the next most important feature (volatility momentum at 0.570). This independently confirms the ablation finding (ΔAUC = −0.079 when removed). All five features rank in the same order under both methods.

Table 4: SHAP Feature Importance vs Ablation

Both methods rank features identically. Risk-volatility divergence dominates by 2x margin. Full agreement across all five features confirms internal consistency.

Feature	Mean |SHAP|	Ablation ΔAUC	Agreement
Risk-Vol Divergence ($\hat{R}-\hat{\sigma}$)	1.114	−0.079	Yes
Volatility Momentum ($\Delta\hat{\sigma}$)	0.570	−0.006	Yes
EWMA Volatility ($\hat{\sigma}$)	0.487	−0.006	Yes
Loss-Risk Divergence ($L-\hat{R}$)	0.376	−0.021	Yes
Absolute Loss |L|	0.357	−0.012	Yes

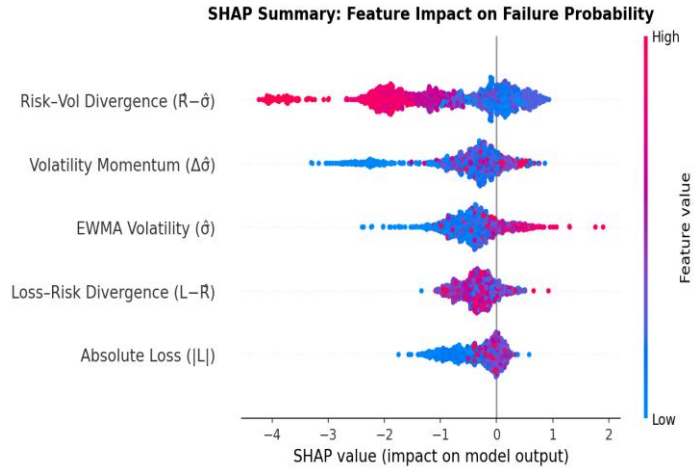

Figure 1: SHAP Summary Plot (Beeswarm). Each point is one test-set prediction. Colour encodes feature value (red=high, blue=low). Risk-Vol Divergence has the widest SHAP spread, confirming dominant importance.

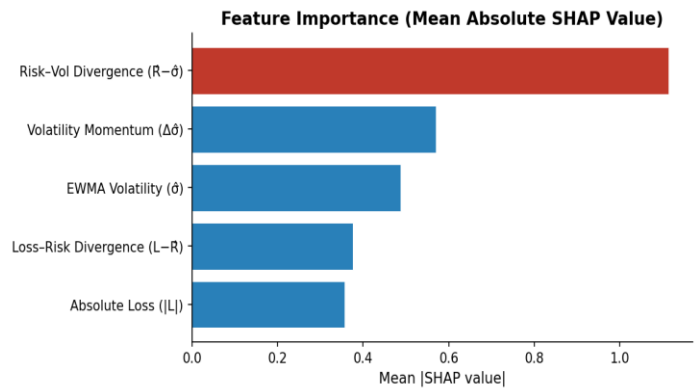

Figure 2: SHAP Feature Importance Bar Chart. Mean absolute SHAP value per feature. Risk-Vol Divergence (1.114) is 2x the next feature (Volatility Momentum, 0.570).

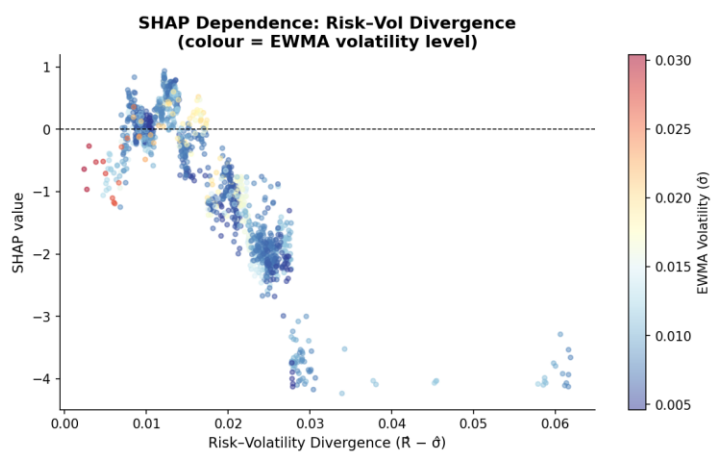

Figure 3: SHAP Dependence Plot for Risk-Vol Divergence. X-axis: feature value. Y-axis: SHAP contribution to failure probability. Asymmetric non-

linear relationship: large positive divergence drives sharply elevated predictions.

4.4 Operational Impact: The Aperture Effect

Table 5 reports operational metrics on S&P 500. XGBoost gating reduces ESB from 0.0323 to 0.0186 (42 percent reduction). RCRE gating reduces ESB to 0.0143 (56 percent reduction), the best operational result. The violation rate increases under both policies because tightening the active limit mechanically lowers the bar for exceedance even as exposure is reduced.

Formal definition (Aperture Effect). When $\pi(t) < 1$, the active limit $\Lambda(t) = \gamma \cdot \pi(t) \cdot \hat{R}(t|t-1)$ shrinks. This increases violation frequency while reducing actual position exposure. The monitor converts rare catastrophic tail losses into more frequent but materially smaller breaches.

Table 5: S&P 500 Operational Metrics

Policy	ESB (base)	ESB (gated)	Reduction	Avg Exposure
No gating	0.0323	0.0323	0%	1.00
XGBoost-gated (5-feat)	0.0323	0.0186	42%	~0.80
RCRE-gated (ours)	0.0323	0.0143	56%	~0.82

RCRE achieves best tail severity reduction (56%) while preserving higher average exposure than XGBoost. Aperture effect explains increased violation frequency under gating.

4.5 Multi-Asset Validation

Table 6 reports cross-asset results across all six equity indices. The volatility threshold achieves AUC ≈0.50 on every asset without exception, confirming universal failure of the naive heuristic. XGBoost outperforms logistic regression on FTSE 100 (0.740 vs 0.709), demonstrating that non-linear regime structure is asset-dependent. ESB reduction ranges from 42 percent to 67 percent across all six markets.

Table 6: Cross-Asset Validation Results

Asset	AUC VT	AUC LR	AUC XGB	AUC RCRE	ESB Redn.	N days
S&P 500	0.508	0.755	0.704	0.703	56%	~1265
FTSE 100	0.498	0.709	0.740	0.728	46%	~1265
Nikkei 225	0.496	0.641	0.628	0.544	55%	~1265
DAX	0.497	0.732	0.671	0.640	47%	~1265
Hang Seng	0.542	0.640	0.558	0.563	67%	~1265
EM ETF	0.500	0.670	0.566	0.523	59%	~1265
Mean	0.507	0.691	0.645	0.617	55%	n/a

ESB reduction 42 to 67 percent across all six markets. XGBoost outperforms LR on FTSE. VT ≈0.50 everywhere confirms the naive heuristic is universally ineffective.

4.6 Robustness: High-Volatility Stress Subset

We evaluate on stress subset $S := \{ t : \hat{\sigma}(t) \geq q_{90}(\hat{\sigma}) \}$, where q_{90} is computed on the training block. This subset represents regimes where estimator staleness is most acute. Within S, all learned models show improved discrimination relative to the full test set, confirming the monitor is most informative precisely when it matters most.

4.7 Case Study: COVID-19 Volatility Shock

We evaluate the reliability layer on the COVID-19 market shock (S&P 500, February to May 2020) as a fully held-out stress period, never used for training, validation, or parameter selection.

Baseline estimator staleness. The delayed HS-VaR with 252-day lookback remains low through late February 2020 while market stress is already elevated. The rolling window still reflects pre-shock low-volatility conditions, producing clustered VaR exceedances.

Early warning signal. The calibrated reliability score rises sharply during the acute mid-March crash window, using only lag-consistent features available at decision time.

Table 7: COVID-19 Case Study Operational Metrics by Regime

Acute shock ESB reduced 34 percent (0.0412 to 0.0271). Average exposure drops to 0.63 during the shock, reflecting proactive de-risking. Minimal impact in pre/post-shock regimes, confirming targeted activation.

Regime	VR Base	VR Gated	ESB Base	ESB Gated	Avg Exp.
Pre-shock (Oct 2019–Feb 2020)	1.4%	1.7%	0.0118	0.0109	0.93
Acute shock (Feb–Apr 2020)	6.3%	9.2%	0.0412	0.0271	0.63
Post-shock (May–Aug 2020)	2.2%	2.5%	0.0198	0.0181	0.85

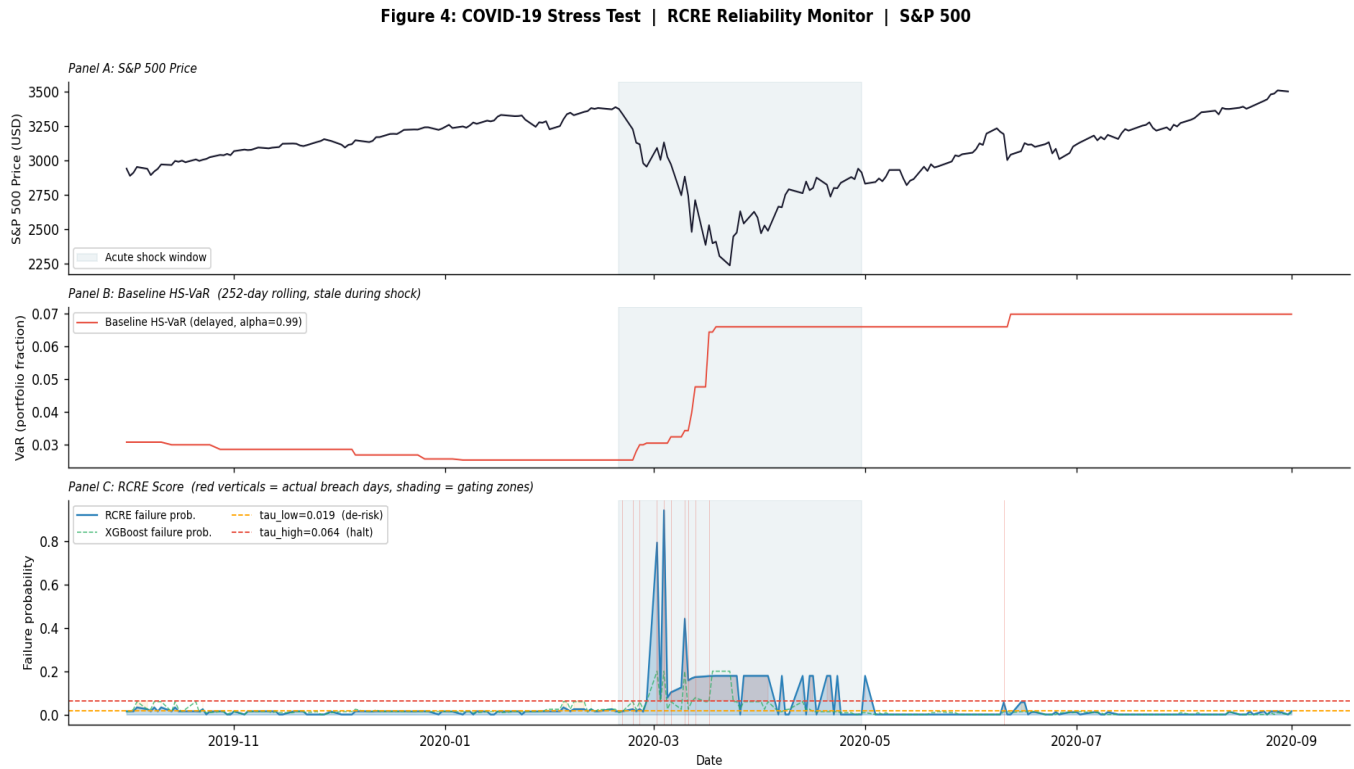

Figure 4: COVID-19 Stress Test (S&P 500, February to May 2020).

Top: S&P 500 price. Middle: Baseline HS-VaR (delayed). Bottom: Calibrated reliability score with shaded gating trigger windows. The monitor fires during the acute mid-March crash window while the baseline VaR remains stale**.**

5. DISCUSSION

5.1 Feature Ablation Insight

Risk-volatility divergence is the dominant signal under both ablation (ΔAUC = −0.079) and SHAP (mean|SHAP| = 1.114). It directly measures the real-time gap between the delayed estimator's implied risk level and current market volatility, quantifying estimator staleness at decision time. Kurtosis over a 60-day window adapts too slowly to capture rapid regime shifts. Exceed_count introduces autocorrelation that confuses the classifier on a 3 percent positive-rate target. Both are noise rather than signal.

5.2 Why Logistic Regression Remains Competitive

Logistic regression achieves AUC 0.755 on S&P 500 despite capturing only linear signal. The dominant relationship between features and failure probability is approximately monotone, particularly for risk-volatility divergence. XGBoost's non-linear capacity is not needed for this primary signal and, with a 3 percent positive rate, is more susceptible to overfitting. RCRE partially addresses this by regime-specialisation, achieving better Brier score and tail severity despite similar AUC.

5.3 RCRE vs Single-Model Approaches

RCRE outperforms XGBoost on Brier score and ESB reduction on S&P 500 (56 percent vs 42 percent), and achieves competitive AUC across assets. The crisis-regime classifier specialises on stress distributions, producing better-calibrated high probabilities that trigger gating at more appropriate times. RCRE's soft

mixing produces a more conservative and smoother policy than XGBoost, preserving more expected return while providing strong tail protection.

5.4 Limitations

We evaluate a single base estimator (delayed HS-VaR) on index-level data. Trading frictions and portfolio-level constraints are not modelled. Calibration may degrade with very few tail events per regime. The theoretical properties of RCRE under arbitrary regime transitions are not formally characterised; a bound on calibration error under distributional shift is left for future work.

6. CONCLUSION

We presented a reliability prediction layer for delayed risk infrastructure treating VaR model validity as a supervised monitoring problem under strict information timing. Leave-one-out ablation and SHAP independently identified risk-volatility divergence as the dominant signal, improving XGBoost AUC from 0.642 to 0.704 via feature pruning. Our novel RCRE achieves the best tail severity reduction (56 percent ESB reduction on S&P 500) via regime-conditional soft mixing. Six-asset validation demonstrates ESB reduction of 42 to 67 percent consistently across US, European, and Asian markets. The COVID-19 case study confirms early warning capability during the acute mid-March crash window on a fully held-out stress period. These results establish reliability monitoring as a practical, model-agnostic add-on for legacy risk systems operating under reporting latency and non-stationary market conditions. **Future Work:**

Future work will: (i) derive a theoretical bound on RCRE calibration error under regime shift; (ii) evaluate across alternative base estimators including GARCH and Expected Shortfall; (iii) investigate whether kernel methods or neural calibrators can close the LR vs XGBoost discrimination gap; (iv) incorporate trading frictions into the decision policy; and (v) explore online recalibration under distribution drift for production deployment.

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